PHYS 143 – Mechanics and Thermal Physics

# Vectors

*How do we describe, combine and calculate with quantities that have both magnitude and direction?*

Hope Donglo

Office 24, Department of Physics, University of Ghana

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

# This week: the central question and the thread

CORE

**Central question**

How do we describe, combine and calculate with quantities that have both magnitude *and* direction?

**By the end of this week, you should be able to:**

1. Represent, add and subtract vectors graphically.
2. Resolve vectors into components and reconstruct their magnitude and direction.
3. Add vectors using components and unit-vector notation, and find the unit vector of any vector.
4. Calculate and interpret dot and cross products.

**The thread of the week**

Vectors give us a way to represent quantities with magnitude and direction. **Graphs build understanding; components give us a systematic way to calculate.**

**1 UNDERSTAND:** why do we need vectors?
**2 CALCULATE:** how do we calculate accurately?
**3 EXTEND:** what can two vectors construct?

**Reading:** Young and Freedman, Chapter 1, §1.7–1.10. **Slide labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

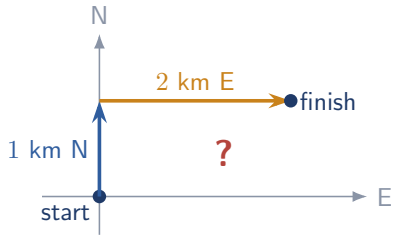
**Opening problem**

You leave the Balme Library and walk **1 km north**, then **2 km east**.

Where are you now, relative to where you started?

**Predict first**

- Are you 3 km from the starting point?
- Is a single number a complete answer?
- In approximately what direction should the answer point?

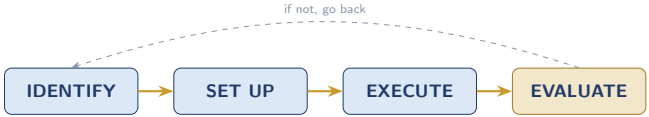



We solve this once we can add vectors at $90^\circ$.

**But what if...** we cannot yet solve this completely. Before solving it, here is the method we will use throughout physics: **IDENTIFY, SET UP, EXECUTE, EVALUATE**.

# How we solve every problem this semester






## IDENTIFY

- Read the problem carefully.
- Identify the target: what is needed?
- Identify the principles of physics required.

## SET UP

- Sketch a diagram. Choose coordinates.
- Look for the appropriate equation.

## EXECUTE

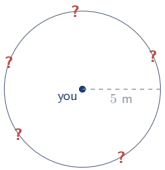
- State the explanation: do the calculation, noting what is missing from what is given.
- Carry units; keep track of signs.

## EVALUATE

- Go over your solution; step back from the paperwork. Are the units, size and direction right?
- Does it make physical sense?

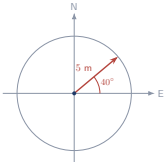
## Contact A: Understand, then start to Calculate

*Why do we need vectors, and how do we calculate with them?*




## Magnitude: how much?

**Define:** the **size** of a quantity: a number with units, such as $5\,\mathrm{m}$, $70\,\mathrm{kg}$, $15\,\mathrm{m/s}$. Never negative. A bag is $5\,\mathrm{m}$ away: which point?




## Direction: which way?

**Define:** the **orientation** of a quantity relative to a reference: a compass word, or an angle (in degrees or radians) from a reference direction.

## Rule: a complete answer to "where?"

**State:** magnitude *and* direction: $5\,\mathrm{m}$ at $40°$ north of east. **Explain:** magnitude alone fits every point on a circle of radius $5\,\mathrm{m}$; the direction picks one.

**But what if...** do all quantities need a direction? Mass is $70\,\mathrm{kg}$ whichever way you face. Some need both, some only a size: **scalars and vectors**.

### Scalar

**State:** a scalar quantity is **described completely by a numerical value and appropriate units**, where applicable.

**Define:** scalars may or may not have units (a count has none; mass has $\mathrm{kg}$). They combine by ordinary arithmetic. e.g. mass, time, temperature, distance, speed, energy.

### Vector

**State:** a vector quantity **requires both magnitude and direction**.

**Define:** the magnitude (a scalar, in the units of the quantity) and the direction (from a reference direction). Vectors combine by the rules of vector addition. e.g. displacement, velocity, acceleration, force.

### Explain: keep these distinct

The magnitude of a vector is a **scalar** and is **never negative**. A vector is not described by a single number: "$15\,\mathrm{m/s}$" is a speed; "$15\,\mathrm{m/s}$ north" is a velocity.

### Notation: vector and magnitude

$$\vec{\boldsymbol{A}} = \text{the vector}$$

$$A = |\vec{\boldsymbol{A}}| = \text{magnitude of the vector}$$

**Define:** a vector is printed bold with an arrow ($\vec{\boldsymbol{A}}$) and handwritten with an arrow ($\vec{A}$). $A$ is its magnitude, a non-negative number in the units of the quantity.

### Convention used all week

In $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = AB\cos\phi$, $A$ and $B$ are magnitudes.
$A_x$ is a **number**; $A_x\hat{\boldsymbol{\imath}}$ is a **vector**.

**Explain:** the arrow says "this is a vector"; dropping it, as in $A$, means "only the size". Mixing the two is the commonest notation error in this course.

# Concept check: scalar or vector?

## Decide for each, then compare with your neighbour

(a) $15\,\mathrm{m/s}$
(b) $15\,\mathrm{m/s}$ north
(c) distance walked, $3\,\mathrm{km}$
(d) displacement, $3\,\mathrm{km}$ east
(e) temperature, $30^{\circ}\mathrm{C}$
(f) a force of $20\,\mathrm{N}$ downwards
(g) $-5\,\mathrm{J}$ of energy
(h) mass, $70\,\mathrm{kg}$

## Answers

Scalars: (a) speed, (c) distance, (e), (g), (h). Vectors: (b) velocity, (d) displacement, (f) force. **A negative number is not automatically a vector:** $-5\,\mathrm{J}$ is a scalar.

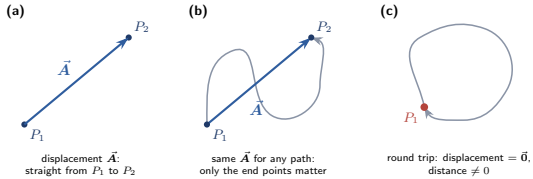
## Which one needs a direction?

A trotro driver says "I am $3\,\mathrm{km}$ from Madina". Is that enough to find the trotro? What is missing? **Distance** is a scalar; **displacement** is a vector (next slide).

**But what if...** how do we describe a change of position as a vector? → **displacement**.

# Displacement: a vector from start to finish

CORE




**Rule: displacement**

**State:** the displacement of a body is the vector from its initial position $P_1$ to its final position $P_2$. It depends only on $P_1$ and $P_2$, **not on the path taken**.

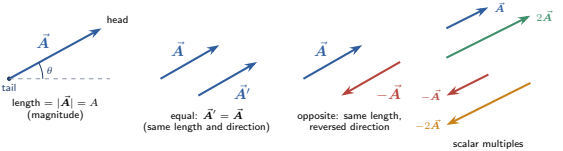
**Define:** magnitude: the straight-line distance $P_1P_2$, in metres; direction: from $P_1$ towards $P_2$.

**Explain:** the distance travelled is the path length, a scalar that can only grow; displacement is zero for a round trip.

**Predict**

You walk once round a $400\,\mathrm{m}$ track and stop where you started. What is the distance? What is the displacement? **Distance $400\,\mathrm{m}$; displacement $\vec{0}$.**

**But what if...** how do we name, draw and compare vectors? → **representing a vector**.




## Rule: equal and opposite vectors

**State:** two vectors are **equal** if they have the same magnitude and the same direction. $-\vec{A}$ has the same magnitude as $\vec{A}$ and the opposite direction.

## Rule: multiplying by a scalar

$c\vec{A}$ has magnitude $|c|A$.
$c > 0$: the direction is unchanged ($2\vec{A}$).
$c < 0$: the direction is reversed ($-\vec{A}$, $-2\vec{A}$).

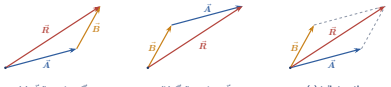
**Define:** a vector is drawn as an arrow (a ray): its length is the magnitude. **Explain:** antiparallel vectors of equal size are negatives of each other.

**But what if...** two displacements follow one another, as in the opening problem. What single displacement results? → **graphical addition**.

# Adding two vectors: three equivalent ways

## Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?




(a) $\vec{A}$ first, then $\vec{B}$ (b) $\vec{B}$ first, then $\vec{A}$ (c) tails together

## Rule: head to tail, (a) and (b)

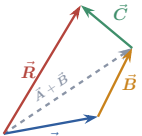
**State:** put the tail of the second vector at the head of the first. The resultant $\vec{R}$ runs from the first tail to the last head. $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ (commutative).

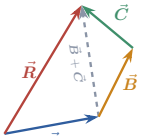
## Rule: parallelogram, (c)

**State:** draw both tails at one point. $\vec{R}$ is the *diagonal* from that point of the parallelogram built on $\vec{A}$ and $\vec{B}$.

**Define:** the **resultant** $\vec{R} = \vec{A} + \vec{B}$ is the single vector with the same effect as $\vec{A}$ then $\vec{B}$. **Explain:** the three constructions give the same $\vec{R}$.

**But what if...** three vectors, not two? Does the grouping matter?



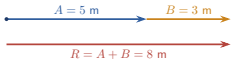

**Rule: associative law**

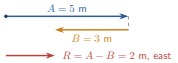
**State:** $(\vec{A}+\vec{B})+\vec{C}=\vec{A}+(\vec{B}+\vec{C})=\vec{B}+(\vec{A}+\vec{C})$: group or order the vectors in any way and the resultant $\vec{R}$ has the same magnitude and direction.

**Define:** $\vec{R}$ is the resultant of all three, from the first tail to the last head. **Explain:** add two, then add the third to that sum; every route ends at the same head. (A numerical check follows in Contact 2.)

**But what if...** we want to undo an addition? First, how big is $\vec{R}$ in the simplest case, along one line.

**Predict first:** sketch $5\,\mathrm{m}$ east then $3\,\mathrm{m}$ east, and $5\,\mathrm{m}$ east then $3\,\mathrm{m}$ west. Predict each resultant.




## Rules: parallel, antiparallel, and the bounds on $R$

**(1) Parallel:** $|\vec{\boldsymbol{R}}| = A + B$, the largest. **Antiparallel:** $|\vec{\boldsymbol{R}}| = |A - B|$, along the *larger* vector, the smallest. In general $|A - B| \leq |\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}| \leq A + B$: for $3\,\mathrm{m}$ and $4\,\mathrm{m}$, $1\,\mathrm{m} \leq R \leq 7\,\mathrm{m}$. **Magnitudes do not simply add.**

## Your turn

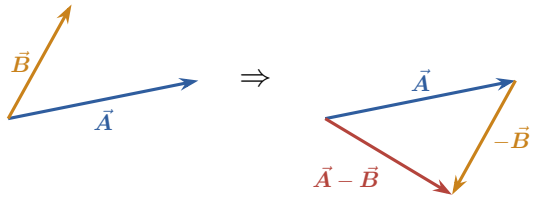
Aayalolo mate walks $4.5\,\mathrm{km}$ east, then (i) $1.5\,\mathrm{km}$ east, (ii) $6.0\,\mathrm{km}$ west. Predict, then find $\vec{\boldsymbol{R}}$. (i) $6.0\,\mathrm{km}$ east; (ii) $1.5\,\mathrm{km}$ west

**But what if...** we want to undo an addition? → **subtraction**.

**Rule: subtraction**

**(2)** $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

Reverse $\vec{B}$, then add head to tail.




**Define:** $\vec{A} - \vec{B}$ is the vector that, added to $\vec{B}$, gives $\vec{A}$. **Explain:** drawn from a shared tail, it runs from the head of $\vec{B}$ to the head of $\vec{A}$.

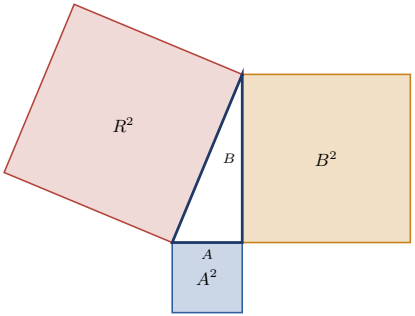
**Connect: Lecture 2**

*Change in velocity* $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$, hence acceleration.

**But what if...** we need numbers, not drawings? A drawing is only as accurate as the ruler. Calculate, starting at $90°$.

**Predict first**

A student walks $5.0\,\mathrm{m}$ east, then $12\,\mathrm{m}$ north across a courtyard. Is the displacement $17\,\mathrm{m}$? Predict its size and direction.




Head to tail, the legs meet at $90°$: a **right triangle**.
**Prediction:** $12 < R < 17\,\mathrm{m}$, leaning north.

**Rule: perpendicular vectors**

**(3)** $R = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1}(B/A)$
**Define:** $A$, $B$ the leg magnitudes; $R$ the resultant (the hypotenuse); $\theta$ the angle between $\vec{\boldsymbol{R}}$ and $\vec{\boldsymbol{A}}$.

$R = \sqrt{5.0^2 + 12^2} = 13\,\mathrm{m}, \quad \theta = \tan^{-1}(12/5.0) = 67°$ north of east. ✓$(12 < 13 < 17)$
**Explain:** valid *only* at $90°$.

**But what if...** does it work when the legs are not at $90°$? First, solve the opening problem.

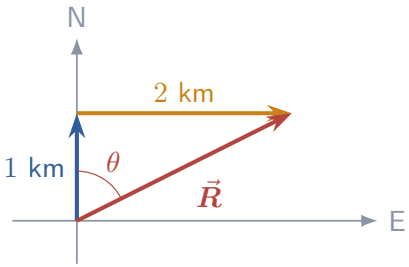
**Example 1**

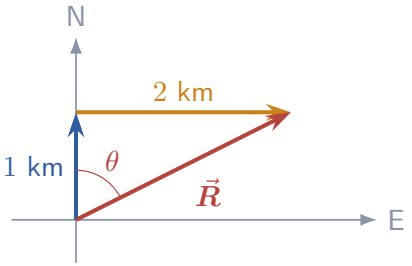
You leave the Balme Library and walk $1.00\,\mathrm{km}$ north, then $2.00\,\mathrm{km}$ east. Find your final displacement from the start: magnitude and direction.

*eg. a walk from the Balme Library, University of Ghana, Legon*

**Predict first**

Is $R$ more or less than $3\,\mathrm{km}$? Closer to north or to east?






**Key idea**

Perpendicular legs make a right triangle: Pythagoras gives the size and the tangent gives the angle.

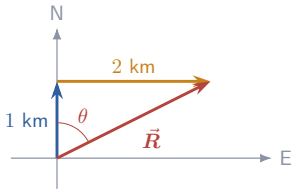
**IDENTIFY**

Given: $A = 1.00\,\text{km}$ (north), $B = 2.00\,\text{km}$ (east), legs at $90°$. Find: $R$ and the direction.

**SET UP**

Head to tail; the legs form a right triangle, so Rule (3): $R = \sqrt{A^2 + B^2}$, $\tan\theta = B/A$ ($\theta$ from north).

Prediction: $2 < R < 3\,\text{km}$, closer to east.




**EXECUTE**

**Write the rule.** $R = \sqrt{A^2 + B^2}$

**Substitute with units.** $R = \sqrt{(1.00\,\text{km})^2 + (2.00\,\text{km})^2}$

**Square and add.** $R = \sqrt{1.00 + 4.00}\,\text{km} = \sqrt{5.00}\,\text{km}$

**Take the root.** $\boxed{R = 2.24\,\text{km}}$

**Write the angle rule.** $\tan\theta = \dfrac{B}{A}$

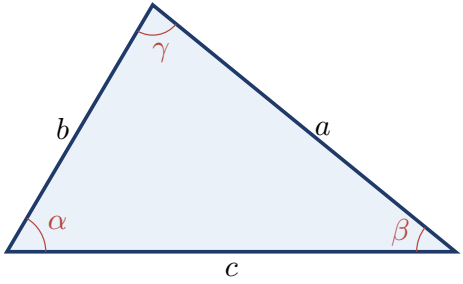
**Substitute.** $\tan\theta = \dfrac{2.00\,\text{km}}{1.00\,\text{km}} = 2.00$ (units cancel)

**Inverse tangent.** $\theta = \tan^{-1}(2.00)$ $\boxed{\theta = 63.4^\circ \text{ east of north}}$

**EVALUATE**

Units km; $2 < R < 3\,\text{km}$ and nearer east, as predicted; $R < 1.00 + 2.00$: magnitudes did not add.

What if the angle between the legs is not $90^\circ$?




side $a$ is opposite angle $\alpha$, etc.

**Rule: cosine rule, find a side**

**(4)** $a^2 = b^2 + c^2 - 2bc\cos\alpha$

For two vectors, $R^2 = A^2 + B^2 - 2AB\cos\gamma$, with $\gamma$ the internal angle opposite $\vec{\boldsymbol{R}}$.

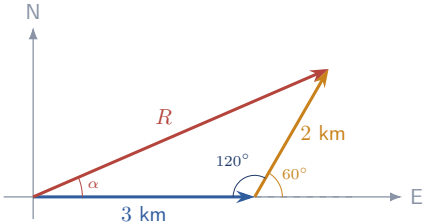
**Rule: sine rule, find an angle**

**(5)** $\dfrac{a}{\sin\alpha} = \dfrac{b}{\sin\beta} = \dfrac{c}{\sin\gamma}$

**Define:** $a$, $b$, $c$ are the sides; $\alpha$, $\beta$, $\gamma$ the angles opposite them. **Explain:** mathematical tools for any triangle, used here without derivation. If $\alpha = 90°$, (4) becomes Pythagoras (3).

**Example 2: a trotro**

A trotro travels $3.00\,\mathrm{km}$ east along the main road, then $2.00\,\mathrm{km}$ at $60^\circ$ north of east along a feeder road. Find the resultant displacement. *Predict first:* is Pythagoras allowed? Is $R$ nearer $5$ or $3\,\mathrm{km}$?



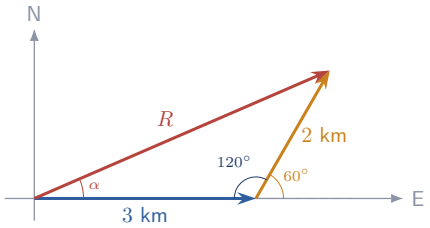

*eg. a trotro route from Madina Station, Accra*

No right angle: the *internal* angle between the legs is $\gamma = 180^\circ - 60^\circ = 120^\circ$. Pythagoras would give $3.61\,\mathrm{km}$, but the second leg alone carries the trotro $2.00\cos 60^\circ = 1.00\,\mathrm{km}$ further east, so $R \geq 4.00\,\mathrm{km}$: $3.61\,\mathrm{km}$ is impossible. Use the cosine rule (4).

**But what if...** we apply Pythagoras to a leg at $60^\circ$? It gives $\sqrt{3.00^2 + 2.00^2} = 3.61\,\mathrm{km}$, yet $R \geq 4.00\,\mathrm{km}$.

# Example 2: a trotro, solution (1): identify and set up

CORE




**Key idea**

The head-to-tail angle between the legs is $120°$, not $60°$, so Pythagoras fails; use the cosine rule with the angle opposite $\vec{R}$.

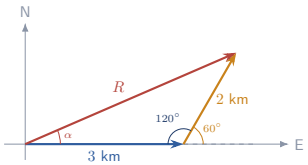
## IDENTIFY

Given: $A = 3.00\,\text{km}$ east, $B = 2.00\,\text{km}$ at $60°$ north of east. Find: $R$ and the direction $\alpha$ above east.

## SET UP

Internal angle opposite $\vec{R}$: $\gamma = 180° - 60° = 120°$. Rules (4) then (5).

Prediction: $3 < R < 5\,\text{km}$, small $\alpha$ above east.




**EXECUTE**

**Write the cosine rule (4).** $R^2 = A^2 + B^2 - 2AB\cos\gamma$

**Substitute with units.**

$$R^2 = (3.00\,\text{km})^2 + (2.00\,\text{km})^2 - 2(3.00\,\text{km})(2.00\,\text{km})\cos 120^\circ$$

**Evaluate each term** ($\cos 120^\circ = -0.500$).

$$R^2 = 9.00 + 4.00 + 6.00\,\text{km}^2$$

**Add, then take the root.** $R^2 = 19.0\,\text{km}^2$ $\boxed{R = 4.36\,\text{km}}$

**Write the sine rule (5).** $\dfrac{\sin\alpha}{B} = \dfrac{\sin\gamma}{R}$, so $\sin\alpha = \dfrac{B\sin\gamma}{R}$

**Substitute.** $\sin\alpha = \dfrac{(2.00\,\text{km})\sin 120^\circ}{4.36\,\text{km}} = 0.397$

**Inverse sine.** $\boxed{\alpha = 23.4^\circ \text{ north of east}}$

**EVALUATE**

$3 < R < 5\,\text{km}$ and $R \geq 4.00\,\text{km}$, as predicted; $0 < \alpha < 60^\circ$, matching the sketch.

**Question**

A delivery rider makes four legs: 2.1 km at $20^\circ$ N of E, 3.4 km at $35^\circ$ W of N, 1.8 km S, 2.6 km at $10^\circ$ S of W. Where does the rider end up?

- Graphical methods give physical understanding . . .
- . . . but scale drawings have limited precision,
- and each extra leg needs another triangle, with its own cosine and sine rule.

**The transition to components**

**Triangle methods work, but become cumbersome. Components give a systematic numerical method for any number of vectors.**

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Giving the magnitude a sign: $\lvert\vec{\boldsymbol{A}}\rvert < 0$. | The magnitude is never negative; the minus sign belongs to the direction. |
| Adding magnitudes: $\lvert\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}\rvert = \lvert\vec{\boldsymbol{A}}\rvert + \lvert\vec{\boldsymbol{B}}\rvert$. | Add head to tail; then $\lvert A - B\rvert \leq \lvert\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}\rvert \leq A + B$. |
| Calling any negative number a vector ($-5\,\mathrm{J}$). | Ask whether a direction is needed: energy is a scalar. |
| Using Pythagoras when the angle is not $90^\circ$. | Find the internal angle, then the cosine and sine rules. |
| Reading "30° north of east" from north. | Start from the first named direction and turn toward the second. |
| Confusing distance with displacement. | Displacement is final minus initial; distance is the path length. |

# Graphs build understanding; components let us calculate

**Where we are:** we can represent vectors and add them graphically and with triangles. **How do we calculate with any number of vectors, at any angles, accurately?**

## Graphical methods

Show what is happening; good for sketches and checks; limited accuracy, awkward for many vectors.

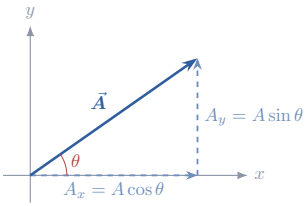
## Components

Turn vectors into ordinary numbers; any number of vectors, any accuracy; the same in 3D.

## The method of the week

**Graphs give geometric understanding; components provide a systematic way to calculate accurately. RESOLVE → COMBINE → RECONSTRUCT**: sketch first, predict, calculate, then check.

**But what if...** we describe a vector by how much of it lies along each axis? What is that "how much"? → **components**.

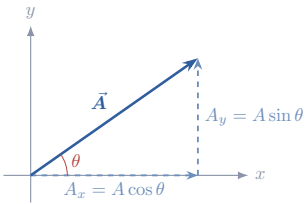



**State.** A 2D vector can be built from an $x$ component and a $y$ component. **Define.** A **component** is the part of a vector along one chosen axis: $A_x$ is **how much of $\vec{A}$ lies along** $x$; $A_y$, **how much along** $y$; units of the vector. **Explain.** A projection on the axis; it depends on the axes chosen.

**Scalar numbers, not vectors**

$A_x$ **is a number** (positive, negative or zero); **the component vector is** $A_x\hat{\boldsymbol{\imath}}$, so $\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{\imath}} + A_y\hat{\boldsymbol{\jmath}}$.

**But what if...** we know only the magnitude $A$ and the angle $\theta$? How do we get $A_x$ and $A_y$? $\rightarrow$ **resolving**.




$\theta$ measured from $+x$ toward $+y$

**Rule: resolving (2D)**

**(6)** $A_x = A\cos\theta, \qquad A_y = A\sin\theta$

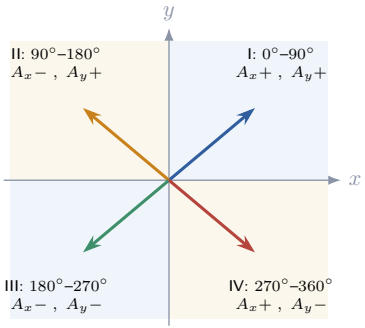
**Apply when $\theta$ is from $+x$ toward $+y$.**

**Define:** $A = |\vec{A}|$; $\theta$ from $+x$ (degrees); $A_x$, $A_y$ numbers in the units of $\vec{A}$. **Explain:** $A_x$ is *adjacent* to $\theta$ ($\cos\theta = A_x/A$), $A_y$ *opposite* ($\sin\theta = A_y/A$).

**Predict first**

For $\theta = 150°$, are $A_x$, $A_y$ positive or negative?

**But what if...** the vector points left or down, as at $150°$? Then $\cos\theta$ or $\sin\theta$ is negative. → **signs and quadrants**.




Before using equations, ask: does the vector point **right or left**? Does it point **up or down**?

**Sign rule**

| quadrant | I | II | III | IV |
|---|---|---|---|---|
| $A_x$ | $>0$ | $<0$ | $<0$ | $>0$ |
| $A_y$ | $>0$ | $>0$ | $<0$ | $<0$ |

**Permanent rule**

**Sketch → predict signs → calculate**

The calculator never replaces the sketch.

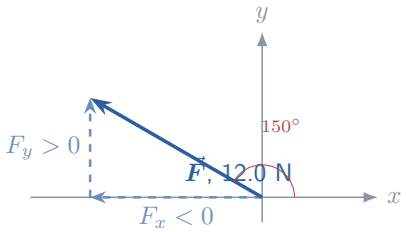
**But what if...** the angle is given from another axis, such as $37^\circ$ from $+y$ or $45^\circ$ below $+x$? Then $A\cos\theta$ and $A\sin\theta$ swap roles. → **resolving carefully**.

**Example 3**

A force of $12.0\,\mathrm{N}$ acts on a wooden stool at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find $F_x$ and $F_y$.

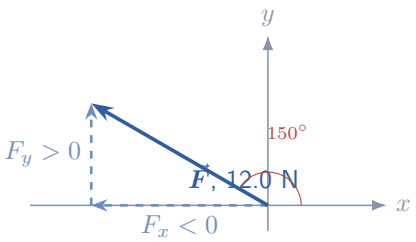
**Sketch → predict signs → calculate**

Right or left? Up or down? So what signs do you expect?



# Example 3: components of a force, solution (1): identify and set up

CORE




**Key idea**

$\theta$ is measured from $+x$, so the signs of $\cos\theta$ and $\sin\theta$ carry the quadrant: sketch first.

## IDENTIFY

Given: $F = 12.0\,\text{N}$, $\theta = 150°$ from $+x$. Find: $F_x$, $F_y$.

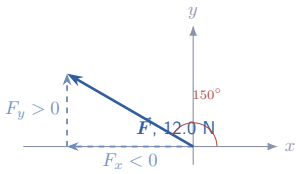
## SET UP

Rule (6): $F_x = F\cos\theta, \quad F_y = F\sin\theta$.

Quadrant II, so predict $F_x < 0$, $F_y > 0$, each smaller than $12.0\,\text{N}$.

# Example 3: components of a force, solution (2): execute and evaluate

CORE




**EXECUTE**

**Write rule (6) for $F_x$.** $F_x = F\cos\theta$

**Substitute with units.** $F_x = (12.0\,\mathrm{N})\cos 150^\circ$

**Cosine is negative in quadrant II.** $F_x = (12.0\,\mathrm{N})(-0.8660)$

$$\boxed{F_x = -10.4\,\mathrm{N}}$$

**Write rule (6) for $F_y$.** $F_y = F\sin\theta$

**Substitute with units.** $F_y = (12.0\,\mathrm{N})\sin 150^\circ$

**Sine is positive in quadrant II.** $F_y = (12.0\,\mathrm{N})(0.5000)$

$$\boxed{F_y = +6.00\,\mathrm{N}}$$

**EVALUATE**

Signs and sizes as predicted. Check: $\sqrt{(10.4)^2 + (6.00)^2}\,\mathrm{N} = \sqrt{108 + 36.0}\,\mathrm{N} = 12.0\,\mathrm{N}$, the given magnitude. ✓

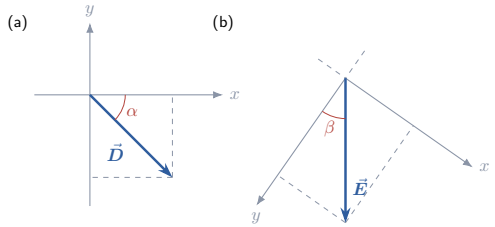
**Example 4**

Find the $x$- and $y$-components of the displacements $\vec{D}$ and $\vec{E}$.

(a) $|\vec{D}| = 3.00\,\mathrm{m}$ of a football; $\alpha = 45.0°$ below $+x$.

(b) $|\vec{E}| = 4.50\,\mathrm{m}$; $\beta = 37.0°$ from the $+y$ axis, with tilted axes.

*eg. a football match at the Accra Sports Stadium*




**Predict first**

Name the axis each angle is measured from. Which of $\sin$, $\cos$ goes with which component? Signs?

**Sketch → predict signs → calculate.**

# Example 4: angles measured from other axes, solution

CORE

**IDENTIFY and SET UP:** rule (6), $A_x = A\cos\theta$, $A_y = A\sin\theta$, holds only for $\theta$ from $+x$; otherwise ask: *adjacent* $\to$ cos, *opposite* $\to$ sin. Predict: $D_x > 0$, $D_y < 0$; $E_x > 0$, $E_y > 0$ (tilted axes).

**(a) Given:** $D = 3.00\,\text{m}$, $\alpha = 45.0^\circ$ below $+x$, so $\theta = -45.0^\circ$.

**Equation (6):** $D_x = D\cos\theta, \quad D_y = D\sin\theta$

**Substitute:** $D_x = (3.00\,\text{m})\cos(-45.0^\circ)$

$D_y = (3.00\,\text{m})\sin(-45.0^\circ)$

**Solve:** $D_x = (3.00\,\text{m})(0.7071)$

$\boxed{D_x = +2.12\,\text{m}}$

$D_y = (3.00\,\text{m})(-0.7071) \quad \boxed{D_y = -2.12\,\text{m}}$

**(b) Given:** $E = 4.50\,\text{m}$, $\beta = 37.0^\circ$ from $+y$.

**Equation:** $E_y$ is *adjacent* to $\beta$, $E_x$ is *opposite*: $E_y = E\cos\beta, \quad E_x = E\sin\beta$

**Substitute:** $E_x = (4.50\,\text{m})\sin 37.0^\circ$

$E_y = (4.50\,\text{m})\cos 37.0^\circ$

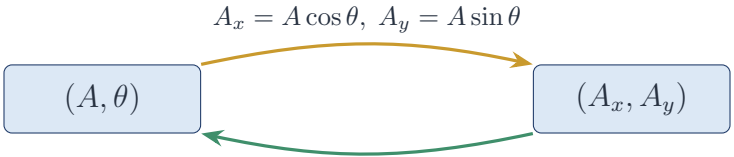
**Solve:** $E_x = (4.50\,\text{m})(0.6018) \quad \boxed{E_x = +2.71\,\text{m}}$

$E_y = (4.50\,\text{m})(0.7986) \quad \boxed{E_y = +3.59\,\text{m}}$

**EVALUATE:** signs as predicted. Check: $\sqrt{2.12^2 + 2.12^2} = 3.00\,\text{m}$ and $\sqrt{2.71^2 + 3.59^2} = 4.50\,\text{m}$, the given magnitudes. ✓ $A\cos\theta$, $A\sin\theta$ **hold only for** $\theta$ **from** $+x$.

**But what if...** given $A_x$ and $A_y$, find $A$ and $\theta$? A calculator gives the same $\tan^{-1}$ for two vectors that point opposite ways. $\to$ **reconstructing**.

# Two equivalent descriptions of one vector



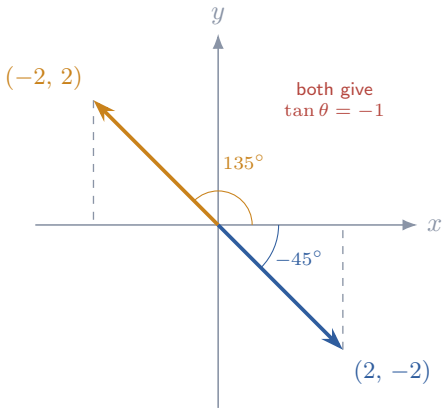

**Key idea**

**Magnitude/direction and components are two equivalent ways of describing the same vector.** Going from components back to magnitude and direction is **reconstructing**.

**Predict first**

$\vec{\boldsymbol{P}} = (-2,\, 2)$ and $\vec{\boldsymbol{Q}} = (2,\, -2)$. Both have $A_y/A_x = -1$, and a calculator gives $\tan^{-1}(-1) = -45°$ for both. Same direction?

**Define:** $\tan\theta = A_y/A_x$, so $\theta = \tan^{-1}(A_y/A_x)$ is a number the calculator returns between $-90°$ and $+90°$ only. **Explain:** two opposite vectors have the same ratio $A_y/A_x$, so the ratio alone cannot give the direction.



**Rule: reconstructing a vector**

**(7) Magnitude:** $A = \sqrt{A_x^2 + A_y^2}$.
**Quadrant:** from the signs of $A_x$, $A_y$.
**Reference angle:**
$\theta_{\text{ref}} = \tan^{-1}|A_y/A_x|$.
**Direction:** I: $\theta_{\text{ref}}$; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

**CHECK:** the direction agrees with the sketch. **Extension:** $\text{atan2}(A_y, A_x)$ builds in both signs; use it after quadrant reasoning. **WATCH OUT:** the same $A_y/A_x$ can mean different directions. **Quadrant first.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.
$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135}^\circ$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.
$\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315}^\circ \ (= -45^\circ)$$

**Your turn:** $A_x = -3.0\,\text{m}$, $A_y = -4.0\,\text{m}$. Find $A$ and $\theta$. $A = \sqrt{(-3.0)^2 + (-4.0)^2} = 5.0\,\text{m}$; quadrant III; $\theta_{\text{ref}} = \tan^{-1}(4/3) = 53.1^\circ$; $\theta = 180^\circ + 53.1^\circ = 233.1^\circ$ ✓

**But what if...** there are several vectors? How do their components combine? → **adding vectors**.

**Contact A: what you can now do**

- Represent a vector and add two graphically, with Pythagoras or the cosine and sine rules.
- Resolve one vector: $A_x = A\cos\theta$, $A_y = A\sin\theta$, signs from the sketch.
- Reconstruct it: $A = \sqrt{A_x^2 + A_y^2}$, quadrant first, then $\theta$.

**Predict first**

**Before Wednesday, predict:** a hiker walks three legs on different compass bearings. If you have the components of each leg, how do you get the total? Why would adding the three angles fail?

**But what if...** we have many vectors to add and want to write them compactly? $\rightarrow$ **adding by components, and unit vectors**.

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

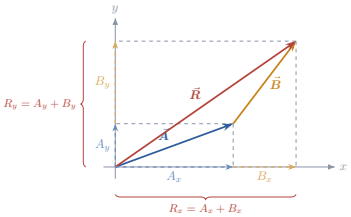
## Contact B: Calculate, then Extend

*How do we add many vectors, and*
*what can two vectors construct?*

**Predict first**

1. A force points $150^\circ$ from $+x$. What are the signs of $F_x$ and $F_y$? $F_x < 0$, $F_y > 0$: second quadrant.
2. A vector has $A_x = -3$ and $A_y = +3$, and the calculator gives $\tan^{-1}(A_y/A_x) = -45^\circ$. What is its direction? $135^\circ$: add $180^\circ$ because $A_x < 0$.
3. Why does the graphical method struggle with four legs? Each extra leg needs another triangle, and scale drawings have limited precision.

# Adding vectors with components




**Question:** how do the components of $\vec{\boldsymbol{R}} = \vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} + \cdots$ relate to theirs?

**Rule: adding vectors by components**

**(8)** $R_x = A_x + B_x + \cdots, \quad R_y = A_y + B_y + \cdots$

**Define:** $R_x$ is the sum of all the $x$-components, and likewise $R_y$ (in 3D also $R_z$). **Explain:** components are signed lengths along one axis, so they simply add.

**The central skill: RESOLVE → COMBINE → RECONSTRUCT**

**RESOLVE** each vector (a table helps), **COMBINE** like components, **RECONSTRUCT** the magnitude and direction of $\vec{\boldsymbol{R}}$. **Sketch → predict signs → calculate**. **CHECK:** the quadrant agrees with the sketch, and $R \leq$ the sum of the magnitudes.

**But what if...** the vectors are given as magnitudes and compass directions? → **a multiple-vector problem**.

**Example 5**

A student on a campus orienteering run makes three displacements:

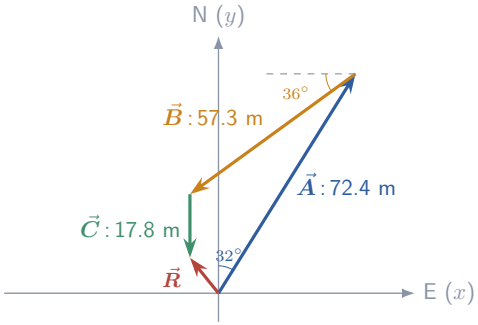
$\vec{A}$: 72.4 m, 32.0° **east of north**
$\vec{B}$: 57.3 m, 36.0° **south of west**
$\vec{C}$: 17.8 m **south**

Find the resultant $\vec{R}$: magnitude and direction.

*eg. the University of Ghana, Legon campus*

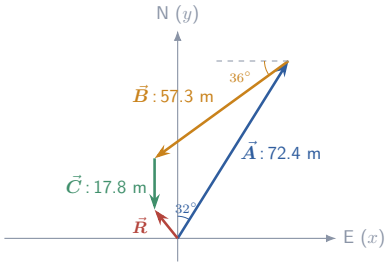
**Predict first**

From the sketch: roughly how far, and in which quadrant?



# Example 5: three displacements, solution (1): identify and set up

CORE




**Key idea**

Convert every angle to one measured from $+x$ so that the same Rule (6) works for all three vectors.

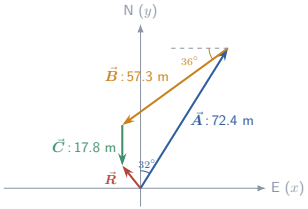
## IDENTIFY

Given: $A = 72.4\,\text{m}$ ($32.0^\circ$ E of N); $B = 57.3\,\text{m}$ ($36.0^\circ$ S of W); $C = 17.8\,\text{m}$ (S). Find: $\vec{\boldsymbol{R}}$.

## SET UP

$x$ east, $y$ north. Angles from $+x$:
$\theta_A = 90.0^\circ - 32.0^\circ = 58.0^\circ$,
$\theta_B = 180.0^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270.0^\circ$.

Rules (6) and (8). Prediction: $R$ much less than the sum of lengths; left and up (quadrant II).

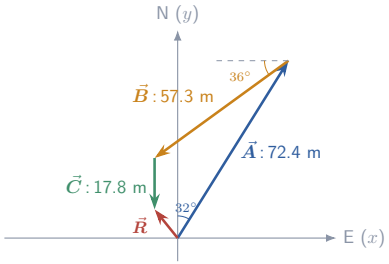



**EXECUTE: resolve each vector**

**Write rule (6).** $A_x = A\cos\theta_A$, $A_y = A\sin\theta_A$; same for $\vec{\boldsymbol{B}}$, $\vec{\boldsymbol{C}}$.

**Substitute with units, one vector per line.**

$$A_x = (72.4\,\text{m})\cos 58.0^\circ = (72.4)(0.5299)\,\text{m} = +38.37\,\text{m}$$
$$A_y = (72.4\,\text{m})\sin 58.0^\circ = (72.4)(0.8480)\,\text{m} = +61.40\,\text{m}$$
$$B_x = (57.3\,\text{m})\cos 216.0^\circ = (57.3)(-0.8090)\,\text{m} = -46.36\,\text{m}$$
$$B_y = (57.3\,\text{m})\sin 216.0^\circ = (57.3)(-0.5878)\,\text{m} = -33.68\,\text{m}$$
$$C_x = (17.8\,\text{m})\cos 270.0^\circ = 0,$$
$$C_y = (17.8\,\text{m})\sin 270.0^\circ = -17.80\,\text{m}$$




**EXECUTE: combine**

**Write rule (8).** $R_x = A_x + B_x + C_x$, $R_y = A_y + B_y + C_y$

**Add the $x$-components.** $R_x = 38.37\,\text{m} - 46.36\,\text{m} + 0$

$\boxed{R_x = -7.99\,\text{m}}$

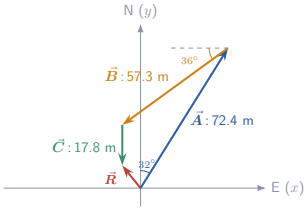
**Add the $y$-components.**

$R_y = 61.40\,\text{m} - 33.68\,\text{m} - 17.80\,\text{m}$ $\boxed{R_y = +9.92\,\text{m}}$

**Read the quadrant from the signs.** $R_x < 0$, $R_y > 0$: quadrant II, as predicted.

**Key idea**

Components add as signed numbers; the quadrant comes from the signs of $R_x$, $R_y$, not from the calculator.




**EXECUTE: reconstruct**

**Write rule (7).** $R = \sqrt{R_x^2 + R_y^2}, \quad \theta_{\text{ref}} = \tan^{-1} \dfrac{|R_y|}{|R_x|}$

**Substitute.** $R = \sqrt{(-7.99\,\text{m})^2 + (9.92\,\text{m})^2}$

**Square and add.** $R = \sqrt{63.8 + 98.4}\,\text{m} = \sqrt{162.2}\,\text{m}$

$\boxed{R = 12.7\,\text{m}}$

**Reference angle.** $\theta_{\text{ref}} = \tan^{-1} \dfrac{9.92\,\text{m}}{7.99\,\text{m}} = 51.1^\circ$

**Direction from $+x$ in quadrant II.** $\theta = 180^\circ - 51.1^\circ = 129^\circ$

$\boxed{\text{i.e. } 39^\circ \text{ west of north}}$

**EVALUATE**

Units m; quadrant II as sketched;
$R = 12.7\,\text{m} \leq 72.4 + 57.3 + 17.8 = 147.5\,\text{m}$, the sum of the magnitudes, because $\vec{\boldsymbol{B}}$ largely cancels $\vec{\boldsymbol{A}}$. ✓

**But what if...** writing components in words is long? → **unit vectors**.

### Unit vectors $\hat{\imath}, \hat{\jmath}, \hat{k}$

**State:** vectors of magnitude $1$, with no units, along the standard directions: $\hat{\imath}$ along $+x$, $\hat{\jmath}$ along $+y$, $\hat{k}$ along $+z$. **Direction only.**

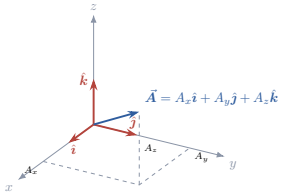
$$\textbf{(9)} \quad \vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

**Define:** $A_x$, $A_y$, $A_z$ are numbers in the units of $\vec{A}$.
**Explain:** a compact way to write components *together with their directions*.

### Number or vector?

$A_x$, $A_y$, $A_z$ are **numbers**; $A_x\hat{\imath}$, $A_y\hat{\jmath}$, $A_z\hat{k}$ are **component vectors**.

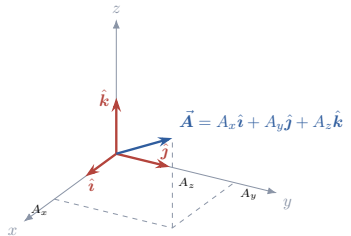
Direction angles in 3D are extended material (supplement).




### Rule: 3D

Right-handed axes: curl the fingers from $+x$ to $+y$, the thumb points along $+z$.

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



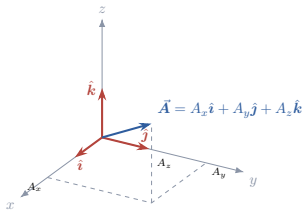

**Key idea**

Add or subtract like components; the unit vectors just label the axis, and $|\vec{F}|$ uses all three squared.

**Example 6**

$\vec{D} = (6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$ and its magnitude.

**IDENTIFY** Given: $\vec{D}$, $\vec{E}$ as above. Find: $\vec{F}$ and $F$. **SET UP** Collect like terms, Rule (8): $F_x = 2D_x - E_x$, and likewise for $y$, $z$; then magnitude in 3D. Predict: $F_y > 0$ since $D_y > 0$ and $E_y < 0$.




**EXECUTE**

**Write the rule.** $F_x = 2D_x - E_x$, and likewise for $y$, $z$

**Substitute $x$.** $F_x = 2(6.00\,\mathrm{m}) - 4.00\,\mathrm{m} = 8.00\,\mathrm{m}$

**Substitute $y$.**

$F_y = 2(3.00\,\mathrm{m}) - (-5.00\,\mathrm{m}) = 6.00 + 5.00 = 11.00\,\mathrm{m}$

**Substitute $z$.**

$F_z = 2(-1.00\,\mathrm{m}) - 8.00\,\mathrm{m} = -2.00 - 8.00 = -10.00\,\mathrm{m}$

**Assemble.** $\boxed{\vec{\boldsymbol{F}} = (8.00\hat{\boldsymbol{\imath}} + 11.00\hat{\boldsymbol{\jmath}} - 10.00\hat{\boldsymbol{k}})\,\mathrm{m}}$

**Write rule (9).** $F = \sqrt{F_x^2 + F_y^2 + F_z^2}$

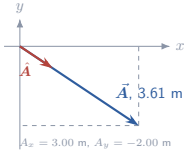
**Square and add.** $F = \sqrt{64.0 + 121 + 100}\,\mathrm{m} = \sqrt{285}\,\mathrm{m}$

$\boxed{F = 16.9\,\mathrm{m}}$

**EVALUATE** $F_y > 0$ as predicted; $3.3\,\mathrm{m} \le F \le 23.8\,\mathrm{m}$ ($|2D| \mp E$). ✓

**Rule: the unit vector of a vector**

**(10)** $\hat{\boldsymbol{A}} = \dfrac{\vec{\boldsymbol{A}}}{A}$ (divide every component by the magnitude), and conversely $\vec{\boldsymbol{A}} = A\,\hat{\boldsymbol{A}}$.




**Define:** $\hat{\boldsymbol{A}}$ has $|\hat{\boldsymbol{A}}| = 1$, no units, and the same direction as $\vec{\boldsymbol{A}}$; $A = |\vec{\boldsymbol{A}}|$. **Explain:** dividing by a positive scalar changes the length, not the direction; so $\hat{\boldsymbol{A}}$ records the direction alone and $A$ the size.

**Predict first**

For $\vec{\boldsymbol{A}} = (3.00\hat{\boldsymbol{\imath}} - 2.00\hat{\boldsymbol{\jmath}})\,\mathrm{m}$: are the components of $\hat{\boldsymbol{A}}$ bigger or smaller than 1? Which signs? Which quadrant?

**But what if...** does the rule give the right numbers? → **a worked example**.

**From combining vectors to constructing quantities**

So far, we have used vectors to represent and combine physical quantities. We now ask a new question: **what useful physical quantities can we construct from two vectors?**

**Pulling a crate**

How much of the pull acts **along the motion**?

pulling a crate $\rightarrow$ **dot product** $\rightarrow$ *work*

vector $\cdot$ vector $\rightarrow$ **scalar**

**Turning a door**

The effect depends on where and which way you push, and has an **axis**.

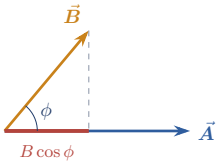
turning a door $\rightarrow$ **cross product** $\rightarrow$ *torque*

vector $\times$ vector $\rightarrow$ **vector**

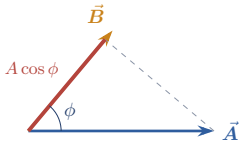
Vector multiplication is carried out in two ways: the scalar ("dot") product and the vector ("cross") product. We first understand what each product means geometrically, then learn how to calculate it.

**Question**

How much of one vector lies along the direction of another?




$\vec{A}\cdot\vec{B} = A\,(B\cos\phi)$




$\vec{A}\cdot\vec{B} = B\,(A\cos\phi)$

**Define.** The part of $\vec{B}$ along $\vec{A}$ is its **projection** $B\cos\phi$; where **for dot and cross products, $\phi$ is the angle between the vectors when placed tail-to-tail**, $0^\circ \le \phi \le 180^\circ$.

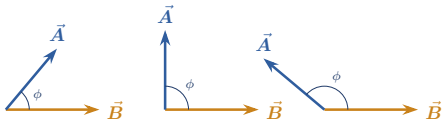
**Definition: dot product**

$$\vec{A}\cdot\vec{B} = AB\cos\phi \quad (11)$$

**Explain.** vector · vector → **scalar**; $A$, $B$ the magnitudes; units multiply: $\mathrm{m}\cdot\mathrm{m} = \mathrm{m}^2$. Equivalently $A(B\cos\phi) = B(A\cos\phi)$: the magnitude of one vector times the component of the other along it.

# Sign of the dot product, and which angle is $\phi$

CORE




(a) $\phi = 50°$ (b) $\phi = 90°$ (c) $\phi = 140°$

**Predict first**

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

**Rule: sign of the dot product**

$$\phi < 90° \Rightarrow \vec{A} \cdot \vec{B} > 0; \qquad \phi = 90° \Rightarrow \vec{A} \cdot \vec{B} = 0; \qquad \phi > 90° \Rightarrow \vec{A} \cdot \vec{B} < 0$$

**Which angle?**

$\phi$ **is the angle between the vectors placed tail to tail.** Do not use the internal angle of a head-to-head triangle: for the trotro legs of Example 2, $\phi = 60°$ but $\gamma = 120°$.

# Dot product from components

**Question**

What are $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}}$?

**Rule: unit-vector dot products**

$$\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}} = 1, \qquad \hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{k}} = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{\imath}} = 0$$

Expanding the product of the two component sums, only like terms survive:

**Rule: dot product from components**

**(12)** $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = A_x B_x + A_y B_y + A_z B_z$

**Define:** $A_x, \ldots, B_z$ are components. **Explain:** (11) uses angles, (12) uses components; both give one scalar.

**Connect: work, Lecture 4**

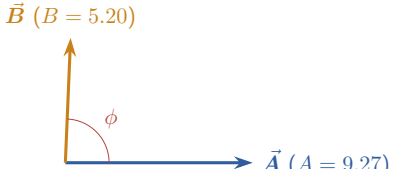
$W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}}$: a constant force $\vec{\boldsymbol{F}}$ acting through a displacement $\vec{\boldsymbol{s}}$. Studied fully in Lecture 4.

**Example 7**

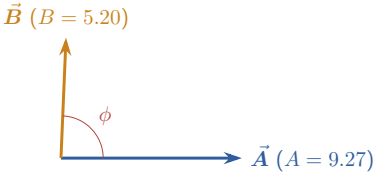
Two vectors are given by $\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$ and $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \cdot \vec{B}$ and the angle $\phi$ between them.

**Predict first**

$\vec{A}$ points mostly along $+y$, $\vec{B}$ mostly along $-x$. Predict: is $\phi$ near $0^\circ$, $90^\circ$ or $180^\circ$?




schematic: tail to tail, $\phi$ is close to $90^\circ$




schematic: tail to tail, $\phi$ is close to $90^\circ$

**Key idea**

Rule (11) solved for $\cos\phi$ turns two vectors into an angle.

**IDENTIFY**

Given: $\vec{A} = (1, 9, 2)$, $\vec{B} = (-5, 1, -1)$. Find: $\vec{A} \cdot \vec{B}$ and the angle $\phi$.

**SET UP**

Rule (12): $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$. Rule (11) solved for the angle: $\cos\phi = \dfrac{\vec{A} \cdot \vec{B}}{AB}$.

Prediction: $\phi$ near $90^\circ$ ($\vec{A}$ mostly $+y$, $\vec{B}$ mostly $-x$).

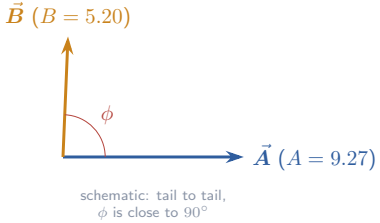
**EXECUTE: the dot product**

**Write rule (12).** $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

**Substitute the components.**

$= (1)(-5) + (9)(1) + (2)(-1)$

**Multiply each pair, then add.** $= -5 + 9 - 2$ $\boxed{\vec{A} \cdot \vec{B} = 2}$




schematic: tail to tail, $\phi$ is close to $90°$

**EXECUTE: the magnitudes**

**Write rule (9).** $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$

**Substitute and add.** $A = \sqrt{1^2 + 9^2 + 2^2} = \sqrt{86}$

$A = 9.27$

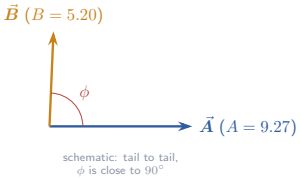
**Same for $\vec{B}$.** $B = \sqrt{(-5)^2 + 1^2 + (-1)^2} = \sqrt{27}$

$B = 5.20$

## Key idea

A small $\cos\phi$ means $\phi$ is just under $90°$, as predicted.

# Example 7, solution (3): angle and evaluate




**EXECUTE: the angle**

**Write the rule.** $\cos\phi = \dfrac{\vec{A}\cdot\vec{B}}{AB}$

**Substitute.** $\cos\phi = \dfrac{2}{(9.27)(5.20)} = \dfrac{2}{48.2} = 0.0415$

**Inverse cosine.** $\phi = \cos^{-1}(0.0415)$ $\quad \boxed{\phi = 87.6^\circ}$

**EVALUATE**

A small positive dot product gives an angle just under $90^\circ$, as predicted.

Independent check in Example 8: $|\vec{A}\times\vec{B}| = AB\sin\phi$.

*Reuse:* magnitude (9), dot product (11) and (12).

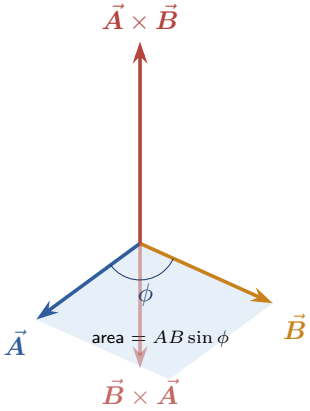
**But what if...** a push perpendicular to a door's lever arm gives $\vec{r}\cdot\vec{F} = 0$, yet turns it most? $\rightarrow$ **cross product**.

**Question**

How can two vectors produce a *third* vector?

**Definition: cross product**

- **(13)** Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$ (area of the parallelogram)
- Direction: perpendicular to the plane of $\vec{A}$ and $\vec{B}$, given by the **right-hand rule**




**Define:** $\phi$ the tail-to-tail angle, $0^\circ \leq \phi \leq 180^\circ$; the result is a **vector**, in the product of the units. **Explain:** zero for parallel vectors, largest for perpendicular ones. Right-hand rule: next slide.

**Right-hand rule**

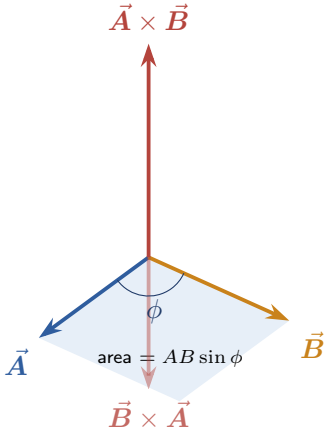
For $\vec{A} \times \vec{B}$:

1. Place $\vec{A}$ and $\vec{B}$ tail to tail.
2. Point the fingers of the right hand along $\vec{A}$.
3. Curl the fingers toward $\vec{B}$ through the smaller angle.
4. The thumb gives the direction of $\vec{A} \times \vec{B}$.

**Order matters**

$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$. Swapping the vectors reverses the direction (same magnitude).

**Special cases:** parallel or antiparallel ($\phi = 0°, 180°$): $\vec{A} \times \vec{B} = \vec{0}$, in particular $\vec{A} \times \vec{A} = \vec{0}$. Perpendicular: $|\vec{A} \times \vec{B}| = AB$, the largest possible.



**Question**

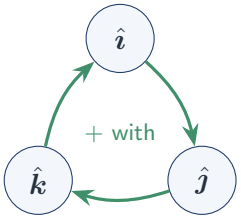
What are $\hat{\boldsymbol{\imath}} \times \hat{\boldsymbol{\jmath}}$, $\hat{\boldsymbol{\jmath}} \times \hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\imath}} \times \hat{\boldsymbol{\imath}}$?

**Rule: unit-vector cross products**

Cyclic order $\hat{\boldsymbol{\imath}} \to \hat{\boldsymbol{\jmath}} \to \hat{\boldsymbol{k}} \to \hat{\boldsymbol{\imath}}$ gives $+$; reversed order gives $-$:

$$\hat{\boldsymbol{\imath}} \times \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}}, \quad \hat{\boldsymbol{\jmath}} \times \hat{\boldsymbol{k}} = \hat{\boldsymbol{\imath}}, \quad \hat{\boldsymbol{k}} \times \hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}}$$
$$\hat{\boldsymbol{\jmath}} \times \hat{\boldsymbol{\imath}} = -\hat{\boldsymbol{k}}, \quad \hat{\boldsymbol{k}} \times \hat{\boldsymbol{\jmath}} = -\hat{\boldsymbol{\imath}}, \quad \hat{\boldsymbol{\imath}} \times \hat{\boldsymbol{k}} = -\hat{\boldsymbol{\jmath}}$$
$$\hat{\boldsymbol{\imath}} \times \hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}} \times \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}} \times \hat{\boldsymbol{k}} = \vec{\mathbf{0}}$$

**Explain:** the three unit vectors are mutually perpendicular, so $\sin\phi = 1$ for unlike pairs and $0$ for like pairs; the sign comes from the right-hand rule.




**Key idea**

Follow the arrows for $+$, go against them for $-$; a product of a unit vector with itself is zero.

**Rule: cross product from components**

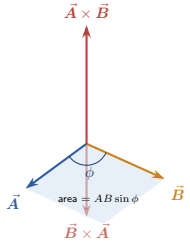
$$\textbf{(14)} \quad \vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

**Define:** the determinant is a memory aid; expand along the top row. For each unit vector, cross out its column and the top row; the $2 \times 2$ left over is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

**$\hat{\boldsymbol{\imath}}$ term:** cross out column 1: $(A_y B_z - A_z B_y)\,\hat{\boldsymbol{\imath}}$.
**$\hat{\boldsymbol{\jmath}}$ term:** cross out column 2: $A_x B_z - A_z B_x$, with a **minus sign**:
$-(A_x B_z - A_z B_x)\,\hat{\boldsymbol{\jmath}} = (A_z B_x - A_x B_z)\,\hat{\boldsymbol{\jmath}}$.
**$\hat{\boldsymbol{k}}$ term:** cross out column 3: $(A_x B_y - A_y B_x)\,\hat{\boldsymbol{k}}$.

**Why the $\hat{\boldsymbol{\jmath}}$ term is written reversed**

The signs along the top row alternate $+, -, +$. Equivalently the coefficients follow the cyclic pattern $yz,\ zx,\ xy$, so the $\hat{\boldsymbol{\jmath}}$ coefficient starts with $A_z B_x$, never $A_x B_z$:
$\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = (A_y B_z - A_z B_y)\hat{\boldsymbol{\imath}} + (A_z B_x - A_x B_z)\hat{\boldsymbol{\jmath}} + (A_x B_y - A_y B_x)\hat{\boldsymbol{k}}$.



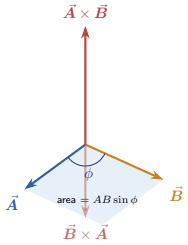

## Key idea

$\vec{A} \times \vec{B}$ is perpendicular to both inputs, so its dot product with each is zero: an independent check.

## Example 8

$\vec{A} = \hat{\imath} + 9\hat{\jmath} + 2\hat{k}$ and $\vec{B} = -5\hat{\imath} + \hat{\jmath} - \hat{k}$. Find $\vec{A} \times \vec{B}$, and check it is perpendicular to both. *Predict: the largest possible magnitude is* $AB = 48.1$.

**IDENTIFY** Given: $\vec{A} = (1, 9, 2)$, $\vec{B} = (-5, 1, -1)$. **SET UP** Rule (14): the determinant, expanded along the top row.




**EXECUTE**

**Write rule (14).** $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

**Expand along the top row, one component per line.**

$x$: $A_y B_z - A_z B_y = (9)(-1) - (2)(1) = -9 - 2 = -11$

$y$: $A_z B_x - A_x B_z = (2)(-5) - (1)(-1) = -10 + 1 = -9$

$z$: $A_x B_y - A_y B_x = (1)(1) - (9)(-5) = 1 + 45 = 46$

**Assemble.** $\boxed{\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -11\hat{\boldsymbol{\imath}} - 9\hat{\boldsymbol{\jmath}} + 46\hat{\boldsymbol{k}}}$

**EVALUATE**

$(\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}) \cdot \vec{\boldsymbol{A}} = (-11)(1) + (-9)(9) + (46)(2) = 0$,
$(\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}) \cdot \vec{\boldsymbol{B}} = (-11)(-5) + (-9)(1) + (46)(-1) = 0$. ✓

Size: $\sqrt{121 + 81 + 2116} = 48.1 = AB \sin\phi$, $\phi = 87.6°$ (Example 7). ✓

# Dot product and cross product compared

## Dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

- Result: a **scalar**
- Zero when the vectors are perpendicular
- Maximum magnitude when the vectors are parallel or antiparallel

## Cross product

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

- Result: a **vector**
- Zero when the vectors are parallel or antiparallel
- Maximum magnitude when the vectors are perpendicular

## The angle $\phi$ in both products

$\phi$ is **the angle between the vectors when placed tail-to-tail**, $0° \leq \phi \leq 180°$. **Do not use the internal angle of a head-to-tail vector triangle as** $\phi$, such as the $\gamma = 120°$ in the trotro problem.

**Connect: torque, Lecture 8 (CORE)**

$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{F}}$, with $|\vec{\boldsymbol{\tau}}| = rF\sin\phi$: a force $\vec{\boldsymbol{F}}$ applied at position $\vec{\boldsymbol{r}}$ from a pivot. The direction comes from the right-hand rule; the order matters; $\tau = 0$ if $\vec{\boldsymbol{F}} \parallel \vec{\boldsymbol{r}}$ and is largest if $\vec{\boldsymbol{F}} \perp \vec{\boldsymbol{r}}$.

**Practice example: a spanner**

A spanner is pivoted at one end; $F = 30\,\mathrm{N}$ acts at $r = 0.50\,\mathrm{m}$ from the pivot at $60^\circ$ to $\vec{\boldsymbol{r}}$ (in the page). **Predict:** is $\tau$ larger or smaller than $rF = 15\,\mathrm{N\,m}$?
$\tau = rF\sin\phi = (0.50)(30)\sin 60^\circ = 13.0\,\mathrm{N\,m}$, less than 15; thumb out of the page, $+z$. ✓

*eg. a mechanic at Suame Magazine, Kumasi*

**But what if...** a problem uses several of these ideas at once: adding, resolving, dot and cross products? Which tool, and when? → **integrated applications**.

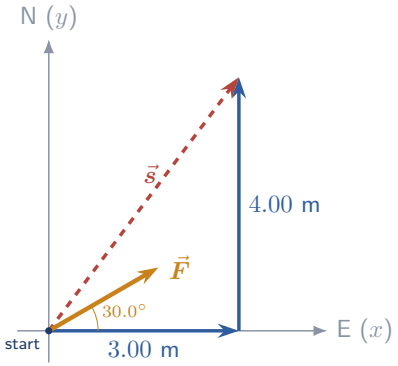
**Example 9: integrated application**

A porter pulls a crate of tomatoes across a level concrete floor with a rope. The rope force is $50.0\,\mathrm{N}$ at $30.0^\circ$ north of east. The crate moves $3.00\,\mathrm{m}$ east, then $4.00\,\mathrm{m}$ north.

(a) Find the crate's displacement $\vec{s}$.
(b) Find the work done by the rope force, $W = \vec{F} \cdot \vec{s}$.
(c) The same force acts at $\vec{r} = (0.800\hat{i} + 0.600\hat{j})\,\mathrm{m}$ from a pivot. Find the torque $\vec{\tau} = \vec{r} \times \vec{F}$.

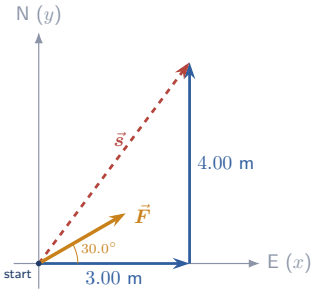
**Predict first**

Sketch first. Which vector tool does each part need, and why? Is $W > 0$? Does $\vec{\tau}$ point along $+z$ or $-z$?




top view; $F = 50.0$ N; $z$ points out of the page

*eg. goods moved at Makola Market, Accra*




top view; $F = 50.0$ N; $z$ points out of the page

**Key idea**

Adding vectors uses components; a scalar from two vectors is the dot product, a vector is the cross product.

## IDENTIFY

Given: $F = 50.0\,\text{N}$ at $30.0^\circ$ N of E; legs $3.00\,\text{m}$ E and $4.00\,\text{m}$ N; $\vec{r} = (0.800\hat{\imath} + 0.600\hat{\jmath})\,\text{m}$. Find: $\vec{s}$, $W$, $\vec{\tau}$.

(a) add vectors → components (8). (b) a scalar from two vectors → dot product (12). (c) a vector from two vectors → cross product (14).

## SET UP

$x$ east, $y$ north, $z$ up. Predict: $\phi < 90^\circ$ so $W > 0$; $\vec{r}$ is steeper than $\vec{F}$, so turning from $\vec{r}$ to $\vec{F}$ is clockwise: $\vec{\tau}$ along $-z$.

## EXECUTE (a)

**Add the legs (8).** $s_x = 3.00\,\text{m}$, $s_y = 4.00\,\text{m}$

$$\boxed{\vec{s} = (3.00\hat{\imath} + 4.00\hat{\jmath})\,\text{m}}$$

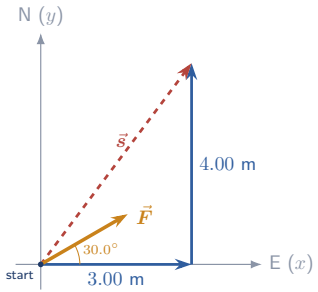
**Magnitude (7).** $s = \sqrt{(3.00)^2 + (4.00)^2}\,\text{m} = 5.00\,\text{m}$

**Direction.** $\theta = \tan^{-1}\dfrac{4.00}{3.00} = 53.1^\circ$ north of east

**Resolve the force (6).** $F_x = (50.0\,\text{N})\cos 30.0^\circ = 43.3\,\text{N}$

$F_y = (50.0\,\text{N})\sin 30.0^\circ = 25.0\,\text{N}$

*Reuse:* (6) resolving, (8) adding.




top view; $F = 50.0$ N; $z$ points out of the page

**Key idea**

Work uses the part of $\vec{F}$ along $\vec{s}$ (dot); torque uses the part perpendicular to $\vec{r}$ (cross, right-hand rule).

**EXECUTE (b): work**

**Write rule (12).** $W = F_x s_x + F_y s_y$

**Substitute.** $W = (43.3\,\text{N})(3.00\,\text{m}) + (25.0\,\text{N})(4.00\,\text{m})$

**Multiply, then add.** $W = 129.9\,\text{J} + 100.0\,\text{J}$ $\boxed{W = 230\,\text{J}}$

**EXECUTE (c): torque**

**Both vectors lie in the $xy$-plane, so only the $\hat{k}$ term of (14) survives.** $\tau_z = r_x F_y - r_y F_x$

**Substitute.** $\tau_z = (0.800\,\text{m})(25.0\,\text{N}) - (0.600\,\text{m})(43.3\,\text{N})$

**Multiply, then subtract.** $\tau_z = 20.0\,\text{N m} - 26.0\,\text{N m}$

$\boxed{\vec{\tau} = -5.98\,\hat{k}\,\text{N m}}$

**EVALUATE**

Work: $\phi = 53.1° - 30.0° = 23.1°$, so $Fs\cos\phi = (50.0)(5.00)\cos 23.1° = 230\,\text{J}$ ✓; $W > 0$ as predicted.

Torque: $r = 1.00\,\text{m}$ at $36.9°$, $\phi = 36.9° - 30.0° = 6.9°$, so $rF\sin\phi = (1.00)(50.0)\sin 6.9° = 6.0\,\text{N m}$ ✓

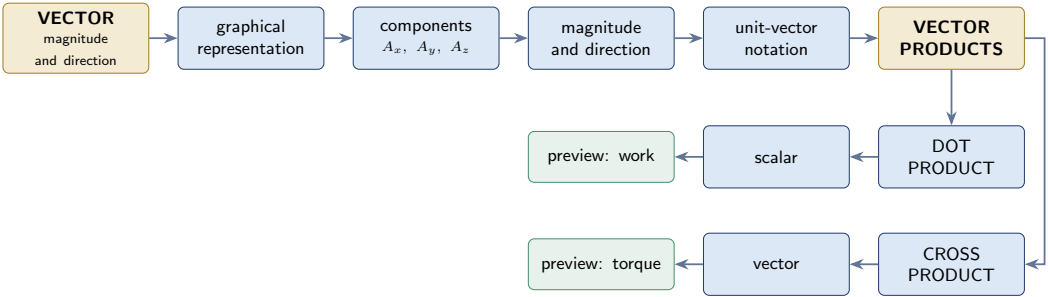
Right-hand rule from $\vec{r}$ $(36.9°)$ to $\vec{F}$ $(30.0°)$: clockwise seen from above, so $-z$ ✓

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Adding magnitudes instead of components. | Resolve, combine, reconstruct. |
| Mixing scalar components with component vectors. | $A_x$ is a number; $A_x\hat{\boldsymbol{\imath}}$ is the vector. |
| Trusting $\tan^{-1}$ for the direction. | Quadrant first from the signs, then the reference angle. |
| Using the internal triangle angle in $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}$ or $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$. | $\phi$ is the angle tail to tail, $0^\circ \leq \phi \leq 180^\circ$. |
| Writing a dot product as a vector, or a cross product as a number. | Dot gives a scalar; cross gives a vector. |
| Forgetting that $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$. | Use the right-hand rule and keep the order. |

# Week 1 concept map

CORE



## Key ideas

- **A vector is a mathematical object used to represent a physical quantity; its components depend on the axes chosen.**
- $A_x$ is a number; $A_x\hat{\imath}$ is the component vector.
- Displacement depends only on the end points, not on the path.
- Magnitudes do not simply add: $|A - B| \leq |\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}| \leq A + B$.
- **Graphs give geometric understanding; components provide a systematic way to calculate accurately.**
- Magnitude/direction and components describe the same vector; find the quadrant before the angle.
- **The dot product measures the component of one vector along another; the cross product produces a vector perpendicular to both input vectors.** Their magnitudes are $AB\cos\phi$ and $AB\sin\phi$, respectively.
- Habit: sketch, predict, calculate, evaluate.

## Reference tools: where to look

**Equations, procedures:** the next slides (master toolkit, core equations).

**Reference deck:** the seven specialised toolkits, equation anatomy, vocabulary, extra examples and practice.

**Common mistakes:** the common-mistakes slide of each contact.

### MASTER TOOLKIT – For a general vector problem

1. **IDENTIFY** What is being asked?
2. **SKETCH** Draw the physical situation.
3. **CHOOSE AXES** Select useful perpendicular coordinates.
4. **PREDICT** Estimate signs, quadrant, size and direction.
5. **RESOLVE** Convert vectors into components if necessary.
6. **COMBINE** Perform the required vector operation.
7. **RECONSTRUCT** Find magnitude and direction if required.
8. **EVALUATE** Check units, signs, quadrant, magnitude, direction and physical reasonableness.

If the result disagrees with the sketch or prediction, investigate before accepting the answer. The seven specialised toolkits are in the Reference deck.

### Inside the course framework

**IDENTIFY** (step 1): what is being asked?
**SET UP** (steps 2–4): sketch, choose axes, predict.
**EXECUTE** (steps 5–7): resolve, combine, reconstruct.
**EVALUATE** (step 8): check the result.

# Reference tools: core equations

| Idea | No. | Equation | Conditions and units |
|---|---|---|---|
| **Along a line** | (1) | $R = A + B$ or $R = \lvert A - B\rvert$; $\lvert A - B\rvert \le R \le A + B$ | one line only |
| **Subtraction** | (2) | $\vec{\boldsymbol{A}} - \vec{\boldsymbol{B}} = \vec{\boldsymbol{A}} + (-\vec{\boldsymbol{B}})$ | same units |
| **Perpendicular** | (3) | $R = \sqrt{A^2 + B^2}, \quad \tan\theta = B/A$ | $90^\circ$ between the vectors |
| **Cosine rule** | (4) | $R^2 = A^2 + B^2 - 2AB\cos\gamma$ | $\gamma$: internal angle opposite $R$ |
| **Sine rule** | (5) | $\dfrac{a}{\sin\alpha} = \dfrac{b}{\sin\beta} = \dfrac{c}{\sin\gamma}$ | sides and opposite angles |
| **Components** | (6) | $A_x = A\cos\theta, \quad A_y = A\sin\theta$ | $\theta$ from $+x$; units of $\vec{\boldsymbol{A}}$ |
| **Magnitude, direction** | (7) | $A = \sqrt{A_x^2 + A_y^2}, \quad \theta_{\text{ref}} = \tan^{-1}\lvert A_y/A_x\rvert$ | place by quadrant |
| **Addition** | (8) | $R_x = A_x + B_x + \cdots = \sum A_x$; likewise $R_y$ | same units for all terms |
| **Unit-vector form** | (9) | $\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{\imath}} + A_y\hat{\boldsymbol{\jmath}} + A_z\hat{\boldsymbol{k}}$ | $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$; $\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}$ no units |
| **Unit vector of $\vec{\boldsymbol{A}}$** | (10) | $\hat{\boldsymbol{A}} = \vec{\boldsymbol{A}}/A, \quad \vec{\boldsymbol{A}} = A\hat{\boldsymbol{A}}$ | $\lvert\hat{\boldsymbol{A}}\rvert = 1$, no units |
| **Dot product** | (11) | $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = AB\cos\phi$ | scalar; product of units |
| **Dot, components** | (12) | $A_xB_x + A_yB_y + A_zB_z$ | same scalar |
| **Cross product** | (13) | $\lvert\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}\rvert = AB\sin\phi$ | vector; right-hand rule |
| **Cross, components** | (14) | $(A_yB_z - A_zB_y)\hat{\boldsymbol{\imath}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{\jmath}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{k}}$ | $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$ |

- ☐ distinguish a scalar from a vector?
- ☐ add and subtract vectors graphically?
- ☐ resolve a vector with the correct signs?
- ☐ determine the direction of a vector, including its correct quadrant?
- ☐ add several vectors by components?
- ☐ write a vector with $\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}$?
- ☐ use IDENTIFY $\rightarrow$ SET UP $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE?
- ☐ calculate a dot product and interpret its sign?
- ☐ calculate a cross product and determine its direction?
- ☐ explain when each product is zero?

**Next**

Problem sheet 1: extra component, resultant, dot and cross problems, plus challenge problems. The Reference deck holds the supporting material cut from the live lecture; the supplementary notes cover linear independence and direction angles in 3D (extension).

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html